England and Wales Price Paid — project pipeline

Stage A — Data preparation

1 Raw register
Price Paid CSV, unchanged
31,702,685 sales

2 Clean panel
positive price, valid date
types D / S / T / F / O

3 Feature matrix
6 z-scored fields
1995–2026, all rows

4 Choose k
Python only, k = 2..8
full file chooses k = 2

Track 1 — Market insight

Purpose: what the sales look like

5 Trends
median price and volume
by type and county

6 k-means groups
two groups of sales
tenure and property type

7 Association rules
type, tenure, price band
Apriori on all cleaned sales

Output — market figures
prices, groups, and rules

Track 2 — Parallel timing

Same steps, timed because the file is large

8 Full-file Lloyd
Python, Spark, YARN, CUDA, GPU Spark
86 rounds, same centres, stop at 1e-4

9 12-piece stitch
Spark, CUDA, and MapReduce
each piece clusters, then one stitch

10 Time the same Apriori
Python, Spark, MapReduce
0.2M rows through the full file

Output — wall-clock seconds
sizes, core counts, and the stitch
one machine, goodpc

Memory check: the 2.22 GB matrix does not fit in a 2 GB limit